

Exploratory Data Analysis & Business Insights Report

Dynamic Delivery Time Prediction System — Statistical Findings

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1. Target Distribution & Summary Statistics

Mean delivery duration across 96,460 orders is **12.56 days** (Median: 10.22 days, Std: 9.55 days, IQR: 6.77 to 15.72 days). The long positive tail extends to 209 days due to exceptional logistics disruptions in remote areas.

2. Correlation & Bivariate Analysis

Feature	Target Correlation	Logistics Interpretation
same_state_delivery	-0.361	Intra-state orders average 7.95d vs 15.13d cross-state.
seller_avg_delivery_days	+0.271	Historically slow sellers consistently generate longer delays.
avg_freight_value	+0.216	Higher shipping cost proxies transit distance.
avg_product_weight_g	+0.081	Heavier packages require slower ground cargo handling.
is_weekend_order	+0.004	Weekend ordering has virtually no impact on total transit time.

3. Operational Takeaways

Intra-state fulfillment reduces transit time by 48%. Implementing merchant dispatch scorecards and dynamic checkout SLAs will protect brand trust and customer retention.